NumPy Lab

Rule: Use NumPy operations; avoid Python loops and lists.

SCENARIO You are analyzing validation results for five AI experiments. Each row is one experiment; the columns are Accuracy, Precision, and Recall.

Starter Data

Create this NumPy array as **results**:

Experiment	Accuracy	Precision	Recall
E1	0.82	0.78	0.80
E2	0.91	0.88	0.90
E3	0.76	0.81	0.79
E4	0.88	0.84	0.86
E5	0.69	0.72	0.70

Part 1 — Understand the Array

1. Print the array and inspect its shape, ndim, size, and dtype. Explain what each value tells you about this dataset.
2. Convert all metric values to percentages (0–100) using one vectorized NumPy operation. Do not write a loop.

Part 2 — Select & Reshape

3. Using NumPy indexing/slicing, extract: (a) E2 Recall, (b) the complete Accuracy column, and (c) the first three experiments with only Accuracy and Precision.
4. Create np.arange(24), reshape it to (4, 6), then flatten it again using ravel(). Confirm the shapes before and after.

Part 3 — Analyze the Results

5. Calculate: the overall mean, the mean of each metric uand the mean score of each experiment using. Explain each result you get.
6. Calculate the standard deviation and variance of the full results array.

Part 4 — Filter & Extend the Dataset

7. Keep only complete experiment rows where Accuracy >= 0.80 AND Recall >= 0.80. Confirm that the filtered result still has three columns.
8. Add a new experiment [0.85, 0.83, 0.87] as a new row.
9. A new metric called F1 is available for all experiments. Create one F1 value per row and add it as a new column. You can initially put all of the F1 values = 0.

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